Driver Drowsiness Detection Using Deep Learning Models Based On Different Camera Positions

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1st Ismail Danish Bin Rozaimi
Vehicle Intelligence and Telematics Lab
School of Electrical Engineering, College of Engineering
Universiti Teknologi MARA, 40450
Shah Alam, Selangor
ismaildanish1998@gmail.com

2nd Fadhlan Hafizhelmi Kamaru Zaman
Vehicle Intelligence and Telematics Lab
School of Electrical Engineering, College of Engineering
Universiti Teknologi MARA, 40450
Shah Alam, Selangor
fadhlan@uitm.edu.my

3rd Syahrul Afzal Che Abdullah
Vehicle Intelligence and Telematics Lab
School of Electrical Engineering, College of Engineering
Universiti Teknologi MARA, 40450
Shah Alam, Selangor
bekabox181343@uitm.edu.my

4th Husna Zainol Abidin
Vehicle Intelligence and Telematics Lab
School of Electrical Engineering, College of Engineering
Universiti Teknologi MARA, 40450
Shah Alam, Selangor
husnaza@uitm.edu.my

5th Lucyantie Mazalan
Vehicle Intelligence and Telematics Lab
School of Electrical Engineering, College of Engineering
Universiti Teknologi MARA, 40450
Shah Alam, Selangor
lucyantie@uitm.edu.my

***Abstract*—Driver drowsiness is among the most critical causes of vehicular incidents. Consequently, a solution to mitigate this problem is exponentially increasing where an intelligent vehicle system is being implemented in vehicle development. In these studies, a solution is developed by implementing Convolutional Neural Network (CNN), which is tasked to identify the driver's drowsiness behaviour. For the datasets that are needed for training and testing of CNN models, datasets are collected at VITAL Laboratories located at UiTM Shah Alam that is recorded from 3 DVR Camera that is placed in the left (CL), middle (CM), and right (CR) positions. The models proposed in these studies are MobileNet-V2, ResNet50, and InceptionV3 architecture, where the performance of each model is evaluated by classifying the behaviour based on three different camera positions. The architecture that is being chosen is InceptionV3 from camera 5, which is placed in the middle positions that successfully achieved an overall accuracy of 85.5% in classifying drowsiness behaviour.**

Keywords—Driver Drowsiness, Convolutional Neural Network (CNN), MobileNet-V2, ResNet50, InceptionV3

I. INTRODUCTION

A vehicular incident transpires when one or more motorized conveyances, such as automobiles and motorcycles, collide with one another. Such incidents can be caused by various factors, such as negligent driving, inattentive driving, adverse weather conditions, or malfunctioning vehicles. Such incidents can result in severe injury or death to passengers or drivers. One of the leading causes of rising car accidents is driver behaviour, particularly drowsy driving, which is a leading contributor to many road accidents in Malaysia. Both nodding and yawning are signs of fatigue or exhaustion, related to sleepiness [1], [2]. Since there has not been enough research done on local drivers to identify tiredness, the issue of driver drowsiness is a significant worry in Malaysia [3]. Despite the numerous deep learning-based strategies currently in use, many of these techniques still need more room for performance improvement. Additionally, which camera position will capture data most effectively to identify drowsy driving has yet to be shown. This problem emphasizes the want for additional study and advancement in this field to enhance the identification and prevention of drowsy driving in Malaysia.

As the issue of driver drowsiness continues to pose a significant threat to road safety, it is crucial to explore and develop practical solutions with the uprising in developing a complete security system in vehicles relevant to the UNESCO Sustainable Development Goals 11, a deep learning approach has been adopted to detect drowsy drivers. A comparison of the Deep CNN Models is being applied and assessed with its performance in predicting driver's distraction using state farm datasets with nine different types of behaviours contrasting to the approach in this study with datasets that are acquired from a recording in a driving simulator with three different types of driver's drowsiness behaviour [4]. In this study, multiple deep learning techniques are being implemented and evaluated for use in embedded systems to detect distracted driving behaviours [5]. The evaluation is conducted using a driving simulator with two cameras monitoring the driver's body and face which is also nearly similar to this study approach. However, the difference is that this study's objective is to utilize three different cameras from different positions that only detect the driver's face

This study aimed to implement a deep convolutional neural network (CNN) and evaluate its performance through model validation and identification of optimal camera positions for detecting driver drowsiness. The study aimed to determine the most effective CNN model for detecting drowsiness by comparing the performance of different models using datasets collected from a laboratory setting. The study also examined the performance of three different DVR camera positions - middle, left, and right - to identify the best camera position for detecting drowsiness. By conducting these evaluations, the study aims to provide conclusions on the best-performing CNN models and camera positions for detecting driver drowsiness.

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II. Related Works

Integration of Internet of Things (IoT) technology with sensor systems to detect and react to possible road dangers is one promising area of research of road safety. According to Chitra et al., one such study investigated using sensors to develop a system that could SMS alert emergency agencies for rapid aid. This study demonstrates how IoT technology can potentially to improve the identification and reaction to accidents and driving while fatigued [6]. A microcontroller's attached sensors can collect data, upload it to the cloud, and store it on the microcontroller itself. The sensors can identify unusual readings in the event of an accident and notify emergency services to get help right away. The system can also be manually activated to notify emergency contacts of your whereabouts. Additionally, a system that uses a fingerprint sensor and a microcontroller to detect the driver's identification was proposed by Almohsen et al. The device also has a vibration sensor built into the car that, in the case of an accident, can instantly alert emergency personnel and pinpoint the exact position of the incident [7]. A proposed system uses a microcontroller equipped with a heart rate sensor for monitoring the driver's heartbeat, and a fingerprint sensor for detecting new drivers, identifying normal heartbeats, and preventing unlicensed drivers from operating a vehicle. The system also has a vibration sensor that can detect any collision. This system uses a GPS and GSM module that relies on a 4G internet connection to communicate with the cloud to collect data on the driver's fingerprint and regular heartbeat. Any changes to the vibration sensor or heart rate will trigger the system to alert emergency services, who will take appropriate action based on the precise location provided by the GPS module. The catch is that it needs an internet connection to function efficiently.

In the field of implementing a camera for an intelligent system vehicle, by using a convolutional recurrent neural network model using data from dashcams, live streaming video recorders, and location data, Takimoto et al. suggested a unique method for predicting the likelihood of accidents [8]. It demonstrates how using DVR camera data could improve accident prediction and prevention. In the research, two models were put out, one incorporating location data and the other not. It was discovered that both models could produce a reliable model for predicting the probability of accidents. The study revealed that the proposed models are more effective at foretelling accidents. The research is constrained, though, because it omits key factors, including information from actual incidents, sensor readings, and map-based data that could improve accuracy. Hayato et al. a solution for bus companies to adjust their routes based on the number of passengers [9]. This solution involves installing a driving recorder, sensors, and a GPS module and using a machine-learning model known as Random Forest regression. The results showed that the system accurately estimates the number of passengers and is also affordable for bus companies to implement. However, the researchers noted room for improvement by incorporating additional information such as acceleration sensor outputs or bus route schedules.

A technique using deep learning architecture to identify driver distractions was proposed by Dipu A. et al. The technique uses a Raspberry Pi 3 and an IP camera to determine closed or open eyes based on a live video stream from the camera, highlighting the potential of deep learning algorithms to improve driver distraction detection and avoidance [10]. Six hundred randomly chosen pictures of yawning, not yawning, open eyes, and closed eyes were used to train a deep learning model. The model underwent testing using the Pascal VOC measure, and it produced accuracy results of 90% under a variety of conditions, including exposure and obstruction. In low light or when the flashlight is directed at the background camera lens, however, it has been noted that the model's detecting function is less reliable, particularly when the device is used for a prolonged time. Vyom et al. also proposed two architectures, InceptionV3 and AlexNet, based on a collection of dashboard images from a State Farm dataset [11]. These architectures produced exceptional results in tracking ten different types of distractions, including using the dashboard, talking on the phone with either hand, interacting with passengers, reaching behind, and drinking. The datasets were used to train two architectures, InceptionV3 and AlexNet, over five epochs. AlexNet obtained an accuracy of 0.9949, while InceptionV3 obtained 0.9928. It was discovered that despite both architectures achieving accuracy levels above 99 percent, AlexNet surpasses InceptionV3. InceptionV3 is used in this paper which is the same as our work, but it differs based on its method, including the purpose of detecting the distraction of the driver

III. Methodology

A. Datasets Preparation

Data Acquisitions and Labelling: The study begins by acquiring datasets of driver drowsiness by conducting experiments with 44 participants in a controlled environment at VITAL Laboratories located in Universiti Teknologi MARA (UiTM) using the setup shown in Fig.1. Due to the concerns for the participant's wellbeing, each participant was asked to perform three types of driver drowsiness behaviours, including normal driving, yawning, and nodding, while seated in a driving simulator to simulate natural driving conditions. The setup for the driving simulator included a steering wheel, a television that was placed in front of the participants, and 3 DVR cameras at three different positions, namely Left (CL), Middle (CM), and Right (CR) set up on the television. Each behaviour was recorded for 60 seconds, and participants were allowed to repeat the actions as many times as desired. Once all participants were recorded, the datasets were acquired and labelled for analysis.



Fig. 1. Driving simulator with the placement camera positions used for data collection.

Data Preparation: The process of retrieving driver drowsiness data involves labelling 44 participants, where a Python code is used to extract images from a video of the

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participants. For datasets training, 80 percent of the data will be distributed, and the remaining 20 percent will be used for testing [12]. In order to do this, datasets are split into training and testing datasets, with 9 participants chosen for testing and 35 participants utilized for training. The image extraction process is then followed by isolating thousands of images of the participants according to the specified behaviours of normal, yawning, and nodding. This process ensures that a large and diverse dataset is available for training and testing the proposed models and algorithms. This is crucial for achieving accurate results in detecting driver drowsiness. The information for the distributions of a dataset is visualized in Table I.

Table I. DISTRIBUTION OF DATASETS (NO OF FILES)

Datasets	*CL*	*CM*	*CR*	*Total of Images*
Training	7,597	7,739	8,028	**23,364**
Testing	2,112	2,014	2,056	**6,182**

Fine Tuning Datasets: "Fine-tuning" a dataset refers to modifying or optimizing it to fit a specific task or model. By utilizing techniques such as data cleaning, data augmentation, and feature selection, the goal is to improve the dataset's quality and make it more suitable for a particular application, such as training a deep learning model. The result of this process is an improvement in the performance of the final model. In this study, participants' datasets were fine-tuned according to their behaviour, where thousands of images were carefully selected to identify the best actions related to yawning and nodding, to increase the accuracy of classifying driver drowsiness.

B. Convolution Neural Network (CNN)

MobileNet-V2: Convolutional neural network MobileNetV2 is created with mobile devices in mind [13]. It is based on an inverted residual structure with residual connections connecting the bottleneck levels. The intermediate expansion layer's source of non-linearity for filtering features is lightweight depth wise convolutions. A 32-filter initial fully convolution layer and 19 additional bottleneck layers are both included in the MobileNetV2 design. Along with that, it employs blocks with a stride of one and two for shrinking. Real-time applications on mobile devices can be used with the MobileNetV2 because of its architecture, which is made to be efficient and light.

ResNet50: ResNet50 is a type of convolutional neural network (CNN) architecture designed to improve the performance of deep CNN models [14]. It is based on the Residual Network (ResNet) architecture designed to overcome the problem of vanishing or exploding gradients in deep CNN models. ResNet50 has a 152-layer architecture and uses skip connections, also known as identity mapping, to combine the results of the stacked layers. This model allows the network to maintain a constant width and height throughout the entire layer. Additionally, the ResNet50 model is simpler and easier to optimize than other networks and is known to converge quickly and produce superior results in computer vision applications such as object identification and face recognition.

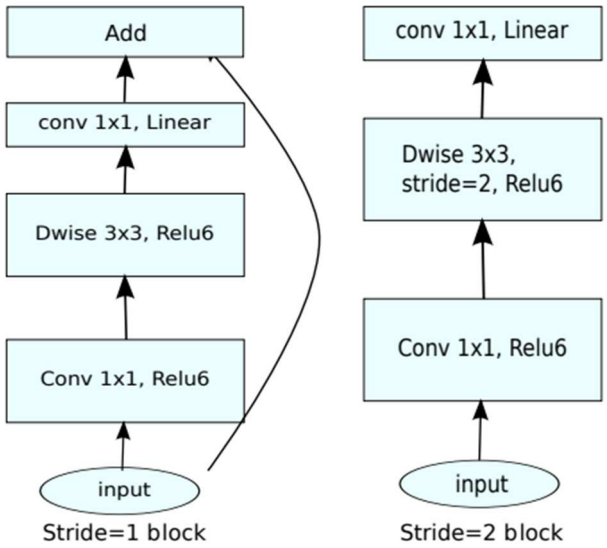


Fig. 2. MobileNet-V2 Architecture [13]

InceptionV3: Specifically designed for image classification problems, InceptionV3 is a convolutional neural network architecture [15]. It is renowned for using computational resources effectively while maintaining high precision. The foundation of InceptionV3 is the Inception architecture, which extracts features from an image by combining convolutional and pooling layers. The network also has a variety of "Inception modules" that use various convolutional filters at various sizes to enable the network to learn features at various levels of granularity. A further method used by InceptionV3 is "batch normalization," which helps to stabilize the training process and enhance performance. It is trained on a sizable dataset like ImageNet and has been utilized in several computer vision applications, including object detection, picture segmentation, and more.

C. Experiments

A custom code is created in MATLAB using our datasets to perform a classification experiment. The datasets are divided into a training set of 35 participants and a testing set of 9 participants, considering factors such as skin colour, ethnicity, wearing a hijab, and the age of the participants. In MATLAB, the CNN model (such as MobileNet-V2) is first set up and the learnable layers of the network are replaced. This step is necessary because the network is pre-trained with Imagenet, which has 1000 output classes. A transfer learning method is employed, where the learnable layers are replaced with our network layers to achieve our desired output classes and are trained using our datasets [16], [17].

To train the datasets, the process is set up with a batch size of 100 and a learning rate of 0.0003. The training process is run for a total of 10 epochs, which means that the model will go through the entire dataset 10 times to learn the patterns and features in the data. Each epoch consists of one forward pass and one backward pass of the entire dataset. It will allow the model to continually update and improve its predictions based on the training data. Additionally, the learning rate is a hyperparameter that controls the step size of the optimization algorithm. It controls how much the model updates during each iteration, a lower learning rate may need more epochs to

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converge, and a higher learning rate may need fewer epochs but may cause an overfitting problem.

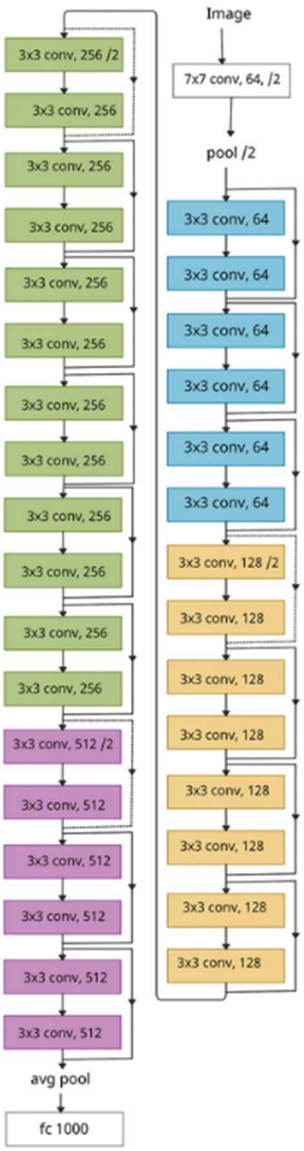


Fig. 3. ResNet50 Architecture [14]

Also, the datasets collected are being augmented to the range from -30 to 30 pixels along with the scale by a factor from 0.9 to 1.1. Data augmentation is a crucial technique where it is artificially creating a new dataset from the existing one where it is being done by adding the noise, rotating, or flipping the original data. The endpoint in creating the augmentation is to create more diverse datasets that can better represent the underlying distribution of the data and to reduce overfitting by creating a more robust model [18].

After completing the training process, the CNN model is saved with the replaced learnable layers. The saved model allows a custom testing code to run on the test datasets. The results of the network's performance can then be evaluated by determining its accuracy and analyzing the confusion matrix. This information can then validate which CNN model has the highest average accuracy and the best confusion matrix for classifying the different behaviours.

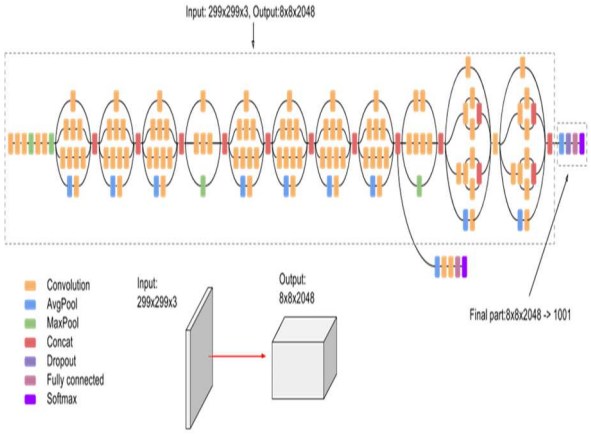


Fig. 4. InceptionV3 Architecture [15]

IV. Results and Discussions

All experiments were conducted in AMD EPYC 24 Core CPU with NVIDIA RTX A5000 GPU. The program was written in MATLAB and the dataset extraction was done with Python 3.9.6. In this section, we will discuss the evaluation of the performance of the CNN Model in classifying driver's drowsiness behaviour with the test datasets with thousands of images from different types of cameras.

Table II below shows the accuracy of three different models in detecting drowsy driving behaviour using three different cameras. The camera in the middle (CM) had the highest accuracy among the three models compared to the cameras on the left (CL) and right (CR). Based on three different models' accuracy from different camera positions suggests that CM is the best camera position for detecting drowsy driving behaviour. The InceptionV3 model had the highest accuracy at 85.5%, which is expected because it is known for maintaining high precision with efficient computing resources. On the other hand, the MobileNet-V2 model had the lowest accuracy at 58.3% and was the worst at detecting drowsy driving behaviour using the CR camera position on the right. It also depicts that the CR camera position is the worst position when comparing the accuracy of these 3 models in detecting the behaviour.

Table II. The Accuracy of 3 Different Model from 3 Different Camera Positions (%)

Models	***CL***	***CM***	***CR***	***Network Parameter***
MobileNet-V2	72.2	**71.3**	58.3	154 layers
ResNet50	62.7	**75.0**	70.4	177 layers
InceptionV3	63.6	**85.5**	66.7	315 layers

The overall performance of the CNN model has limitations in classifying the driver's behaviour, particularly distinguishing between normal driving and nodding behaviours as some pictures may appear similar. Fine-tuning the dataset can help mitigate this issue by filtering out pictures of **"Nodding"** behaviour to ensure they genuinely depict nodding. This filtering process applies to all cameras, such as CL, CM, and CR. However, even with fine-tuning, the

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model's overall accuracy often only reaches 70% due to inconsistencies in deleting the dataset and the way the camera positions can affect our ability to accurately classify the image. Fig.5 illustrates the confidence when classifying the behaviour of driver drowsiness using some of the testing datasets from three different camera positions.

Fig. 5 highlights the challenges faced by the CNN model (InceptionV3) in accurately identifying driver drowsiness behaviour based on its confidence rate. The example presented is split into two sections, the left side showcases a poor classification, and the right side displays a correct classification. The data displayed in the figure was taken from CM and showed a scenario where the model incorrectly classified the driver's behaviour as **"Normal Driving"** with a confidence rate of 38.7% when the driver displayed **"Yawning"** behaviour. This issue also occurred with the other two cameras, CL, and CR, where InceptionV3 mistook the behaviour as **"Yawning"** and **"Normal Driving"** when to the naked eye, it was clearly **"Nodding"** behaviour.

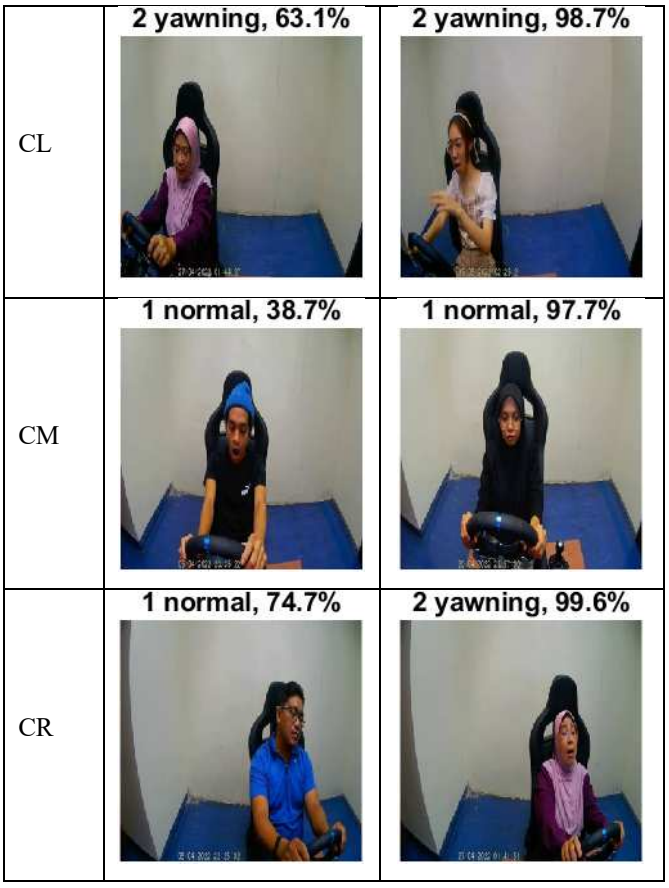
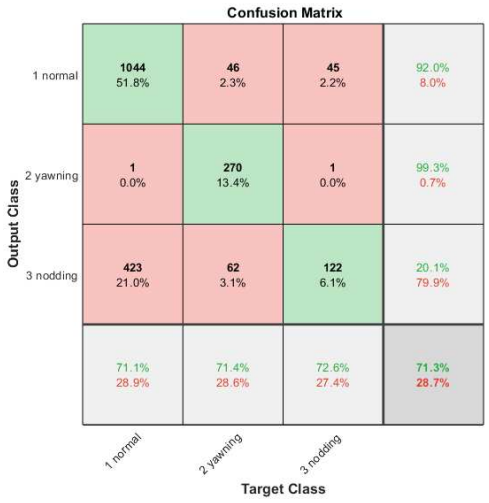
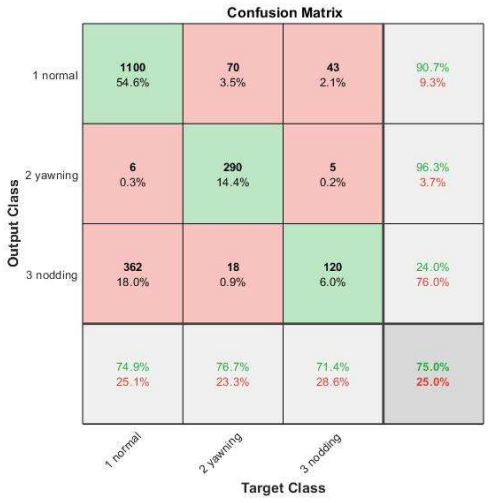


Fig. 5. Confidence rate of detected behaviors from Left (CL), Middle (CM), and Right camera (CR)

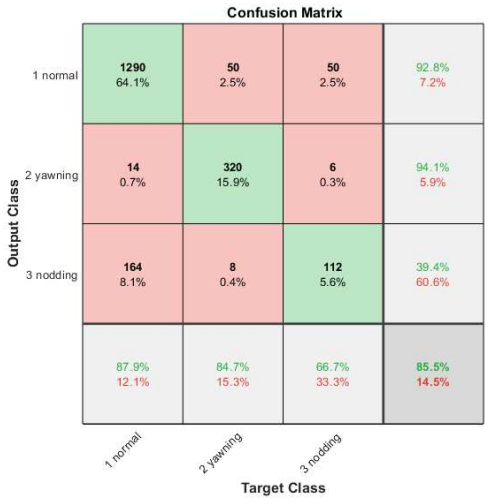
Fig.7 displays the confusion matrix results for three CNN models when applied to data captured by the best camera position (CM), which is located in the middle of the driving simulator setup. The confusion matrix shows the challenges faced in the study due to significant confusion between **"Normal driving"** behaviour and **"Nodding"** behaviour, as the driver's head positions can appear similar in these two actions.



(a)



(b)



(c)

Fig. 6. Confusion Matrix of (a) MobileNet-V2, (b) ResNet50 and (c) InceptionV3 for CM camera position

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All three models showed that the nodding datasets were mainly categorized as normal behaviour, with MobileNet-V2 and Resnet50 misclassifying most of the datasets as normal driving behaviour. In contrast, InceptionV3 only misclassified 164 in the lower right box. These three CNN models' confusion matrices highlight the challenge of accurately identifying the behaviour due to limitations during the study, such as inconsistencies during the fine-tuning process and the similar head orientation between nodding and normal driving behaviour.

Table III shows the accuracy of 3 different models in detecting driver behaviour using a camera that positions in the middle (CM). InceptionV3 had the best overall performance in identifying behaviour, with 87.9% accuracy for normal driving and 84.7% for yawning but only 66.7% for nodding. However, MobileNet-V2 had the highest accuracy in detecting nodding behaviour, with an impressive 72.6%. Overall, InceptionV3 is the best model for detecting behaviour with the highest accuracy, while MobileNet-V2 is the best for classifying nodding behaviour compared to the other two models.

Table III. THE ACCURACY OF 3 DIFFERENT MODEL FROM MIDDLE CAMERA POSITIONS FOR DROWSY DRIVING BEHAVIOUR (%)

Models	***Normal***	***Yawning***	***Nodding***
MobileNet-V2	71.1	71.4	**72.6**
ResNet50	74.9	76.7	71.4
InceptionV3	**87.9**	**84.7**	66.7

V. CONCLUSIONS

InceptionV3 is ideal in choosing the CNN Models with CM producing a result with the highest accuracy in classifying drowsy driver behaviours after analysing the performance of other CNN models. It achieved an accuracy of 87.9% for **"Normal Driving"**, 84.7% for **"Yawning"** and 66.7% for **"Nodding"**. The utilization of InceptionV3 shows that the method of classifying driver drowsiness may be further developed into a complete system that may be deployed in a car and recognize the behaviours in real-world simulation in a controlled environment. It is important to note that there are some restrictions on how the behaviour may be classified, as InceptionV3 might find it challenging to distinguish between **"Normal Driving"** and **"Nodding"** behaviour because the driver's head orientations may appear similar in images. Moreover, the accuracy of the model in detecting the behaviour is influenced by the camera's position. In future research, the collected datasets from CM can be utilized in conjunction with InceptionV3 to identify driver drowsiness in real-life simulations, addressing one of the limitations of this study.

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